Tutorial 4: Instrumentation Filters

Based on Lecture 4 - Instrumentation Filters

Purpose: This tutorial follows the supplied Lecture 4 and the style of the earlier EE3400 tutorials. It covers filter types, Bode plots, cut-off frequency, passive RC filters, second-order filters, band-pass/RLC filters and active filters.

Instructions: Show all working for numerical questions. Use SI units unless otherwise stated. Explain what numerical results mean physically.

Useful formula reference

Relationship	Formula
Capacitor reactance	Xc = 1/(2 pi f C)
Inductor reactance	XL = 2 pi f L
First-order RC cut-off	fc = 1/(2 pi R C)
Second-order RC cut-off	fc = 1/[2 pi sqrt(R1 R2 C1 C2)]
Bandwidth	BW = fH - fL
Centre frequency	fr = sqrt(fH fL)
RLC resonant frequency	fr = 1/(2 pi sqrt(L C))
Quality factor	Q = fr/BW
Active non-inverting gain	Av = 1 + R2/R1

A. Filter Fundamentals

1. Define the **pass band** of an analogue filter. What happens to frequencies outside the pass band?

2. Differentiate between a **passive filter** and an **active filter**. State one advantage and one limitation of each.

3. Explain the function of: (a) low-pass, (b) high-pass, (c) band-pass, and (d) band-stop/notch filters.

4. A low-pass filter has a cut-off frequency of 500 Hz. Classify 50 Hz, 250 Hz, 500 Hz and 2 kHz as pass-band, cut-off or attenuated frequencies.

B. Bode Plots and Filter Response

5. What is a Bode plot used to show? Identify the quantities represented on its axes.

6. Explain the meaning of the corner/cut-off frequency and the -3 dB point. What voltage-response magnitude corresponds to the -3 dB point according to the lecture?

7. A filter has a pass-band gain of 10 V/V. Calculate the gain at the cut-off frequency if the response there is 0.707 of the pass-band value.

8. A first-order low-pass filter has a cut-off frequency of 1 kHz. What attenuation slope is expected in the stop band?

9. A third-order low-pass filter is formed from three first-order sections. Determine its ideal attenuation slope.

10. What is pass-band ripple? Why is a relatively flat pass band useful in measurement systems?

C. Passive RC Filters

11. For an RC low-pass filter, explain how capacitor reactance changes with frequency and why this produces low-pass behaviour.

12. Design a first-order RC low-pass filter with R = 10 kOhm and f_c = 100 Hz. Calculate C.

13. A first-order RC high-pass filter has R = 4.7 kOhm and C = 0.1 microF. Calculate f_c.

14. A first-order RC low-pass filter has R = 2.2 kOhm and C = 0.47 microF. Calculate f_c.

15. For Question 14, state whether 10 Hz, 150 Hz and 2 kHz are mainly passed or attenuated.

D. Second-Order Filters

16. Explain what is meant by the order of a filter. What happens to attenuation slope when first-order sections are cascaded?

17. A second-order low-pass filter has R1 = R2 = 10 kOhm and C1 = C2 = 100 nF. Calculate f_c using the second-order equation in the lecture.

18. Compare the expected roll-off of first-order and second-order filters and explain the difference.

E. Band-Pass and RLC Filters

19. A band-pass filter has f_L = 200 Hz and f_H = 2 kHz. Calculate bandwidth and centre frequency.

20. An RLC series band-pass filter has L = 10 mH and C = 1 microF. Calculate resonant frequency.

21. Explain what happens at resonance in a series RLC band-pass filter and why the output is maximum around resonance.

22. An RLC band-pass filter has centre frequency 1 kHz and bandwidth 100 Hz. Calculate Q.

F. Active Filters

23. Why is an operational amplifier used in an active filter? State two benefits of active filters.

24. An active low-pass filter uses a non-inverting amplifier with R1 = 10 kOhm and R2 = 40 kOhm. Calculate its pass-band/DC gain.

25. An active high-pass filter has R1 = 5 kOhm and R2 = 20 kOhm in the amplifier section. Calculate the gain.

26. A second-order active low-pass filter has R3 = R4 = 10 kOhm and C1 = C2 = 10 nF. Calculate f_c. If R1 = 10 kOhm and R2 = 20 kOhm, calculate amplifier gain.

27. Explain the roles of C1 and C2 in a second-order active low-pass filter and why adding the second RC section produces a steeper attenuation slope.

28. An active band-pass filter is formed using a high-pass stage, amplification stage and low-pass stage. Explain why the two filter sections define the final pass band.

G. Integrated Engineering Problems

29. A sensor signal contains useful frequencies from DC to 100 Hz and unwanted high-frequency noise above 1 kHz. Which filter would you choose and why? Suggest a suitable cut-off region.

30. A measurement system must reject DC and low-frequency drift, pass 200 Hz to 2 kHz, and attenuate frequencies above 2 kHz. Identify the filter arrangement. Calculate bandwidth and centre frequency.

31. Discuss the advantages and disadvantages of placing an amplifier before a low-pass filter. Is filtering always required before amplification?

32. Design a conceptual third-order high-pass filter using the lecture's approach. State the number of first-order sections and the ideal attenuation slope.

Student Checklist

[ ] Identify pass band and stop band for each filter type.

[ ] Explain the -3 dB/cut-off point.

[ ] Calculate Xc, XL and RC cut-off frequency.

[ ] Distinguish first-order and second-order roll-off.

[ ] Calculate bandwidth, centre frequency and Q.

[ ] Explain resonance in a series RLC filter.

[ ] Calculate active-filter gain.

[ ] Explain how cascading sections changes attenuation rate.

[ ] Explain the roles of the RC sections and op-amp in active filters.